

2 Vector Functions

2.2 Calculus with Vector Functions

Definition 2.2.1 If $\vec{r}(t)$ is a vector function, its derivative is defined by

$$\vec{r}'(t) = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}.$$

Provided that this limit exists.

Theorem 2.2.2 If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, and f , g , and h are differentiable, then

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle.$$

Definition 2.2.3 The unit vector with the same direction as the tangent vector to $\vec{r}(t)$ is called the unit tangent vector and is given by

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}.$$

Example 2.2.4 Find the derivative of $\vec{r}(t) = \langle 1 + t^3, te^{-t}, \sin 2t \rangle$, and find the unit tangent vector at the point $t = 0$.

Theorem 2.2.5 Suppose $\vec{u}$ and $\vec{v}$ are differentiable vector valued functions, and c is a scalar, and $f(t)$ is a real valued function, then

- $\frac{d}{dt}(\vec{u} + \vec{v}) = \vec{u}'(t) + \vec{v}'(t)$
- $\frac{d}{dt}(c\vec{u}(t)) = c\vec{u}'(t)$
- $\frac{d}{dt}(f(t)\vec{u}(t)) = f(t)\vec{u}'(t) + f'(t)\vec{u}(t)$
- $\frac{d}{dt}(\vec{u}(t) \cdot \vec{v}(t)) = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$
- $\frac{d}{dt}(\vec{u}(t) \times \vec{v}(t)) = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$
- $\frac{d}{dt}(\vec{u}(f(t))) = f'(t)\vec{u}'(f(t))$

Theorem 2.2.6 If $|\vec{r}'(t)| = c$ is a constant for all t , then $\vec{r}''(t)$ is orthogonal to $\vec{r}'(t)$ for all t .

Example 2.2.7 Compute the derivative of the vector valued function defined by

$$\vec{r}(t) = \sin(t) \cdot \langle 1, t, e^t \rangle.$$

The integral of a vector valued function is similarly computed: If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, and each of f , g , and h is integrable on $[a, b]$, then

$$\int_a^b \vec{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle.$$

Example 2.2.8 Compute the indefinite integral of the vector valued function

$$\vec{r}(t) = \langle 2 \cos t, \sin t, 2t \rangle.$$

2.3 Arc Length

Definition 2.3.1 Given a vector valued function $\vec{r}(t)$, defined on an interval $[a, b]$, the length of the curve $\vec{r}(t)$ from a to b is given by

$$L = \int_a^b |\vec{r}'(t)| dt.$$

Example 2.3.2 Find the length of the curve $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$ from $a = 0$ to $b = 2\pi$.

2.3.1 The Arc Length Function

Definition 2.3.3 Let $\vec{r}(t)$ be a vector valued function, defined on $[a, b]$, we can define it's arc length function as

$$s(t) = \int_a^t |\vec{r}'(u)| du.$$

Example 2.3.4 Reparametrize the helix $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$ with respect to arc length measured from the point $t = 0$.